COMP34612: Computational Game Theory

Group Project (Coursework)

Overview

Your task is to create a program that plays a repeated 2-person Stackelberg pricing games as the leader, under conditions of imperfect information.

Followers' Specifications

- There are three different followers, named as MK1, MK2, and MK3. In addition, there are three hidden followers, MK4, MK5, and MK6. Please refer to Section Performance of the Approach for more information.
- The payoff function and strategy spaces of these followers are unknown. However, historical data from 100 days of previous games between the leader and each follower is available and provided.

Leader's Specifications

Assume that the leader's strategy (i.e., price) is u_L and the follower's strategy (i.e., price) is u_F. The specifics regarding the leader's strategy and profits are as follows:

- **Unit cost**: $c_L = £1.00$
- **Strategy space**:
    1. When playing with MK1, MK2, MK4, and MK5, the leader's strategy space is $U_L = [1.00, +\infty)$
    2. When playing with MK3 and MK6, the leader's strategy space is $U_L = [1.00, 15.00]$
- **Demand Model** (i.e., price-sale relationship): $S_L(u_L, u_F) = 100 - 5u_L + 3u_F$
- **Daily Profit**: $(u_L - c_L)S_L(u_L, u_F)$
- **Objective**: The leader's objective of playing is to maximise the accumulated profit for the next 30 days.
- **Assumption**: For simplicity, it is assumed that the leader's unit cost, strategy space, and the demand model remain unchanged throughout the entire period of 130 days.

Game Rules

On day t, the leader announces the price, $u_L(t)$, first. After learning the leader's price, the follower will choose the responding price, $u_F(t)$. This process occurs over a period of $t = 1, 2, \ldots, 130$, where $t = 1, 2, \ldots, 100$ corresponds to the days for which historical data is available, and $t = 101, 102, \ldots, 130$ are the days during which the game will be played repeatedly.

Game Playing Scenarios

In this setup, your program, acting as the leader, will engage in three separate games against three different competitors, identified as followers MK1, MK2, and MK3.

For each of the repeated game, the leader selects his learning method to learn the follower's reaction function based on the provided set of historical data which spans from $t = 1, 2, \ldots, 100$. During the next 30 days ($t = 101, 102, \ldots, 130$), the leader announces a price, $u_L(t)$, to the game platform daily. After the leader's price is announced, the follower will respond with the price, $u_F(t)$. Upon receiving the follower's price, $u_F(t)$, from the game platform, the leader may use this new information to refine their knowledge/ update the learned reaction function and decide his price of the next day. In the other words, the leader (i.e., your program) should be able to take the follower's price of the previous day from the game platform for updating, and then decide and send his price of the next day to the game platform.

It should be emphasised that the follower's strategy and payoff function is subject to the changing and time varying environment. That is, the parameters in the follower's payoff function may not be the same every day.

Implementation Options

1. **Single Leader Approach**: You may choose to design and implement a single leader algorithm that competes against all three followers, adapting its strategy based on the opponent being faced.
2. **Multiple Leaders Approach**: Alternatively, you could design and implement three distinct leaders, each tailored to play against one of the followers. This approach allows for specialised strategies to be developed for each follower type.

Whichever method you choose, it's crucial to document in your submission which leader(s) is designed to play against each follower. This clarity will help in understanding the design choices and strategies implemented in your program. Please see below for more information.

Project Timeline

- **Week 7 (Starting 16 March):** Start to work on your project and project planning.
- **Week 11 (4 May – 8 May):** Group presentations telling us what you have done.
- **Deadline (8 May at 14:00):** Deadline to submit codes and supporting document.

Support

- **Weekly Support Sessions**: Project support sessions are scheduled every Tuesday at 9:00 am between Week 7 (17 March) and 10 (28 April). These sessions will be

held at Chemistry_G.54. You are encouraged to attend if you have questions, need clarification, or seek guidance on any aspect of the project. Attendance is optional and based on your needs for support.

- **Email**: In addition to the weekly sessions, you are welcome to ask questions or seek support via email at any time. If you encounter any issues with installing or running the game platform, or if you have any other project-related inquiries, do not hesitate to contact the teaching assistants by email for support.

Assessment Criteria

The project is a significant component of your final grade, accounting for 50% of the total mark, with the remaining 50% coming from the final examination. The project mark is distributed as follows:

1. **Content of the Approach (50% of this project mark)**:
   - Presentation (40%): Your grade in this category is based on the presentation you give in Week 11. This includes the originality of your idea, how well it integrates knowledge from the course, and any relevant external literature or research you have incorporated.
   - Written Materials (10%): This is based on the documentation you submit to your group journal on Blackboard. It evaluates the depth and clarity of your written explanation of the approach, including the rationale behind your strategies and methods.
2. **Performance of the Approach (50% of this project mark)**:
   - This evaluates the effectiveness of your submitted program by looking at how well it performs in the game platform, that is, the accumulated profit generated by your submitted codes.
3. **Group Mark Distribution**:
   - The overall group mark will be allocated to individual group members based on self-assessment and the demonstration of knowledge during the presentation.

Presentation

During the presentation week of Semester 2, every group is scheduled to present their project to the lecturers. It is important for each group member to actively participate in speaking and answering questions. This participation allows us to evaluate each member's contributions to the project. The assessment criteria focus solely on the content of your presentation, including the activities your group undertook and the lessons learned through the project process. Presentation skills are not part of the evaluation, so the emphasis should not be on creating a polished presentation but on effectively communicating your project's substance and outcomes.

Performance of the Approach – Assessment Details

- **Step 1.** We will run your submitted codes against MK1, MK2, and MK3 and get

the accumulated profit for the days 101 – 130. These profits will be used as the performance of your submitted codes against MK1, MK2 and MK3

- **Step 2**. We have also designed 3 further followers MK4, MK5, and MK6, which are created by slightly changes of the parameters of MK1, MK2, and MK3 respectively That is, MK1 and MK4 are very similar, MK2 and MK5 are very similar, MK3 and MK6 are very similar. No data about MK4, MK5, and MK6 is provided. That is, they are invisible followers to you. But the setting of each MK4, MK5, and MK6 is the same as the provided followers, with 100 historical data and we will run your codes to get the accumulated profit for days 101 – 130, which will be used as the performance of your submitted codes against MK4, MK5 and MK6.
- **Step 3.** Based on the accumulated profits for MK1 and MK4 and compared with the best possible performances of MK1 and MK4, your codes will get a percentage mark; Similarly, you will get a percentage mark from MK2 and MK5, and then a percentage mark from MK3 and MK6. Finally, your performance mark will be the average of these 3 percentage marks.

Remark. The motivation and justification for the creation of MK4, MK5, and MK6 are as follows:

- We have run the codes submitted by the project groups of the last year to MK4, MK5, and MK6, and noticed various issues, such as some submissions were hard coded for some parameters rather than via the learning; Some were invalid such as out of the price bounds; Some have the running errors etc.
- In real applications such as petrol station pricing, it is impossible to many experiments and tests for each station, but we need a piece of software usable across a large numbers of petrol stations. The creation of MK4, MK5, and MK6 is to simulate and test such a need from a real application point of view.
- In the last year, many groups produced very similar performance due to the limited test cases. The creation of MK4, MK5, and MK6 will provide more testing cases and help distinguishing the performance of different approaches.
- As MK4, MK5, and Mk6 are invisible to you, this should not add your workload for the project.

Submission

Please note that

- There is an **execution time limit of 10 minutes per run**, and all submissions will be **marked using the free version of Google Colab**.
- **No baseline will be provided**; this is intended to encourage independent problem formulation and to promote original thinking, rather than reliance on a predefined solution approach.

Before submitting, please make sure that:

- Your implementation works within this environment.
- Avoid using any external dependencies that might not be available.

Please download your notebook for submission by clicking "File" in the top left corner, then selecting "Download" followed by "Download .ipynb". If your notebook requires any external files to run, ensure that you submit them along with your notebook. Submission will be done through Canvas, and further details will be provided later.

Game Platform and Resources

The project will be run on Google Colab, which is a hosted Jupyter Notebook service. This is done to avoid compatibility issues across different operating systems and to provide a collaborative environment where group members can work effectively using Colab's sharing functionality.

Setting up Google Colab

1. Download Project Files

   You will need two files for this project, ensure both files are downloaded to your local machine.
   - comp34612_project.ipynb (The notebook file)
   - comp34612.zip (The project files)
2. Upload Project notebook
   - Open Google Colab by clicking this link: Google Colab
   - Click "New notebook" in the top right.
   - From the top menu, go to File → "Upload notebook".
   - Select and upload comp34612_ project.ipynb.
   - If a prompt pops up, click "Leave" to proceed to comp34612_ project.ipynb notebook.
3. Connect to a Runtime
   - After uploading the notebook, check if it is connected to a runtime.
   - In the top-right corner, you should see a green checkmark with RAM and Disk information. If it does not show, click "Connect" to establish a connection.
4. Upload the Project Files

   You need to upload the zip file to Colab environment. Please do NOT unzip it.
   - From the left sidebar, click on the folder icon to open the file browser.
   - Upload the python files
     - Option 1: Drag and drop the file.
     - Option 2: Click the upload button (below "Files") and select the file.
   - If a warning prompt will pop up, click "OK" to proceed.

Using Google Colab for the Project

⇒ Running Code
- Click on a cell and press Shift + Enter, or
- Click the Play button on the left side of the cell

⇒ Adding Cell
- Hover over an existing cell and click "+ Code" or "+ Text" where needed

⇒ Stopping a Running Cell
- From the top menu, click “Runtime” → “Interrupt execution”.

⇒ Restarting Runtime
- Click “Runtime” → “Restart session”.
- This will clear all variables.

⇒ Disconnecting and Deleting the Runtime
- Click “Runtime” → “Disconnect and delete session”.
- This will reset the Colab environment completely.

⇒ Sharing with Group Members
- You can share your notebook with your group members by clicking “Share” in the top-right corner.
- Be careful when saving, if multiple people edit the notebook at the same time, you may overwrite each other’s work.

⇒ For more functionalities, please visit this link.

Understanding the Notebook Structure

The notebook is structured into several sections. You can click the first icon on the top left to open the “Table of contents” and navigate through different sections.

1. Project Description: This section provides an overview of the project.
2. Install and Import: This section installs, imports necessary packages, and unzips comp34612.zip.
3. Base Leader and Example Leader: This section contains the base leader class and an example leader implementation.
4. Your Leader: In this section,
   - First write your group number as an integer.
   - Then implement your leader(s) based on the project requirements.
   - Clearly document:
     - Which implementation approach you choose (single or multiple).
     - Which leader is designed to play against each follower.
5. Simulation: This section provides an interface to simulate the pricing game between your selected leader and selected follower.